

■ Grade 6 STEM — Electrochemistry Lesson Plan

Subject: Science / STEM

Grade Level: 6

Duration: 60 minutes

■ Lesson Overview

In this hands-on lesson, Grade 6 students discover how chemical reactions between two different metals produce electricity. By building a simple galvanic cell, learners see firsthand the real-world science behind everyday batteries, developing both conceptual understanding and practical lab skills.

■ Learning Objectives

- Define electrochemistry and explain why it matters in daily life.
- Identify and describe the components of a galvanic cell (anode, cathode, electrolyte, salt bridge).
- Explain how electrons flow through an electrochemical circuit.
- Build a working galvanic cell using copper, zinc, and a saltwater electrolyte.
- Connect galvanic cell principles to real-world applications such as batteries and electric vehicles.

■ Key Vocabulary

Term	Definition
Electrochemistry	The branch of science studying chemical reactions that produce or use electricity.
Galvanic Cell	An electrochemical cell that converts chemical energy into electrical energy spontaneously.
Anode	The negative electrode where oxidation (loss of electrons) occurs.
Cathode	The positive electrode where reduction (gain of electrons) occurs.
Electrolyte	A liquid or gel that allows ions to move, completing the internal circuit.

Salt Bridge	A device (e.g., saltwater-soaked paper towel) that allows ions to flow between two half-cells.
Electron Flow	The movement of electrons from the anode to the cathode through an external wire.

■ Materials Required

- Copper strip (or thick copper wire)
- Zinc strip (or galvanised nail)
- Saltwater solution (2 tbsp salt in 1 cup warm water)
- OR fresh lemon juice as alternative electrolyte
- Two small containers / cups
- Paper towel soaked in saltwater (salt bridge)
- Alligator-clip wires (×2)
- LED bulb (3–5 V)
- Safety gloves (per student)
- Goggles (per student)

■■ Lesson Procedure

Part 1 — Introduction (10 min)

- Engage:** Begin with the question: "Where does the energy in your phone battery come from?" Collect answers; introduce the term electrochemistry.
- Explain:** Describe a galvanic cell as a tiny battery — two different metals joined by a liquid that lets electricity flow. Use the 'water-slide for electrons' analogy.
- Show:** Display or sketch a diagram of a galvanic cell: anode (Zn) → wire → cathode (Cu), with an electrolyte and salt bridge.

Part 2 — Guided Build (30 min)

- Step 1 — Electrolyte:** Dissolve 2 tbsp of salt in a cup of warm water and stir until clear.
- Step 2 — Set up electrodes:** Place the copper strip on one side of the container and the zinc strip on the other. Ensure they do NOT touch.
- Step 3 — Connect wires:** Clip one alligator wire to the copper strip and one to the zinc strip. Connect the free ends to the LED bulb.
- Step 4 — Add the salt bridge:** Soak a paper towel in saltwater; drape it across both containers to connect the two electrolyte solutions.

**Step 5 —
Observe:**

The LED should glow! Students record observations — brightness, any changes over time, bubble formation on the zinc.

Part 3 — Discussion & Real-World Links (10 min)

- Discuss:** Why did the LED light up? Trace the electron path from Zn → wire → LED → Cu → through the electrolyte → salt bridge → back to Zn.
- Connect:** Everyday batteries, electric cars (rechargeable galvanic cells), digital clocks, flashlights, and remote controls all use this same principle.
- Ask:** "What would happen if we swapped the metals? Or used vinegar instead of saltwater?"

Part 4 — Assessment / Quiz (10 min)

- Partner Quiz:** Students answer the three quiz questions with a partner before sharing with the class.
- Debrief:** Clarify misconceptions; summarise key concepts from the lesson recap slide.

■ Safety Guidelines

■ Wear Safety Gloves

Always wear gloves when handling metals and chemical solutions.

■■ Goggles On

Wear goggles throughout the experiment to protect your eyes.

■■■ Adult Supervision Required

Work only with a teacher or parent present.

■ No Tasting or Drinking

Never taste or drink any solutions used in the lab.

■ Wash Hands After

Wash hands thoroughly with soap and water when finished.

⇒ ■ Knowledge Check — Quiz Questions

Students answer individually, then discuss with a partner before sharing answers with the class.

Q What two metals did we use in our galvanic cell, and which one is the anode?
1.

Answer: _____

Q What is the role of the electrolyte in the galvanic cell?
2.

Answer: _____

Q Why do we need a salt bridge? What happens if it is removed?
3.

Answer: _____

Q Name two real-world devices that rely on galvanic cell technology.
4.

Answer: _____

Q In your own words, describe the path electrons take from the anode to the cathode.
5.

Answer: _____

■ Lesson Recap

■ Chemical reactions create electricity.

Electrochemistry demonstrates that interactions between different metals and an electrolyte can generate a usable electric current.

■ Galvanic cells are tiny batteries.

We built our own battery: electrons flow from the zinc anode → external wire → LED → copper cathode. Ions move through the electrolyte and salt bridge to complete the circuit.

■ Science is everywhere.

AA batteries, car batteries, EV power packs, clocks, and flashlights all rely on the same galvanic cell principles students explored today.

■ Safety is non-negotiable.

Gloves, goggles, adult supervision, no tasting, and hand-washing protect everyone in the lab.